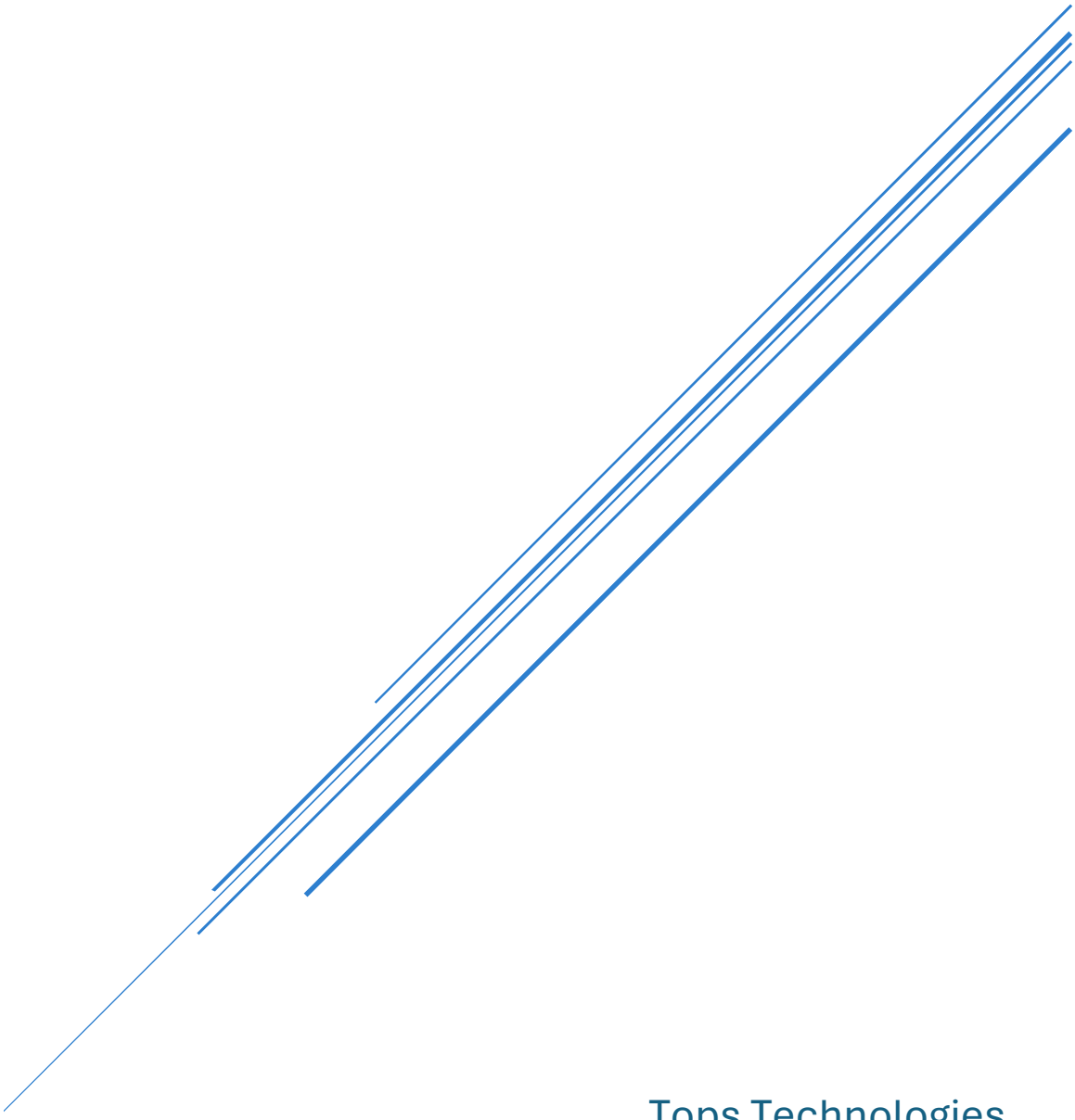
CONFIGURE IPV6 ADDRESSING

Assignment - 8



Tops Technologies
Cyber Security

1. **Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).**

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```
Microsoft Windows [Version 10.0.26300.9711]
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C:\Windows\System32>netsh interface ipv6 show interfaces

Idx     Met         MTU          State                Name
---  ----------  ----------  ------------  ---------------------------
  1          75  4294967295  connected     Loopback Pseudo-Interface 1
 17          30        1500  connected     Wi-Fi
 15           5        1500  disconnected  Ethernet
 13          25        1500  disconnected  Local Area Connection* 1
 21          25        1500  disconnected  Local Area Connection* 2
 12          35        1500  connected     VMware Network Adapter VMnet1
  3          35        1500  connected     VMware Network Adapter VMnet8


C:\Windows\System32>netsh interface ipv6 add address interface="Wi-Fi" address=2001:db8:1::10
The object already exists.


C:\Windows\System32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

```
thernet adapter Ethernet:

  Media State . . . . . . . . . . . : Media disconnected
  Connection-specific DNS Suffix  . :

ireless LAN adapter Local Area Connection* 1:

  Media State . . . . . . . . . . . : Media disconnected
  Connection-specific DNS Suffix  . :

ireless LAN adapter Local Area Connection* 2:

  Media State . . . . . . . . . . . : Media disconnected
  Connection-specific DNS Suffix  . :

ireless LAN adapter Wi-Fi:

  Connection-specific DNS Suffix  . :
  IPv6 Address. . . . . . . . . . . : 2001:db8:1::10
  IPv6 Address. . . . . . . . . . . : 2401:4900:8f72:fe76:4e43:a89:cd49:8ffd
  Temporary IPv6 Address. . . . . . : 2401:4900:8f72:fe76:8425:b1b8:f5b7:31e3
  Link-local IPv6 Address . . . . . : fe80::91d1:6f4e:b9fe:919b%17
  IPv4 Address. . . . . . . . . . . : 192.168.1.8
  Subnet Mask . . . . . . . . . . . : 255.255.255.0
  Default Gateway . . . . . . . . . : fe80::5a07:f8ff:fe85:caf1%17
                                      192.168.1.1

thernet adapter VMware Network Adapter VMnet1:

  Connection-specific DNS Suffix  . :
```

2. **Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.**
   Hint: Use the ping6 or ping -6 command for IPv6 testing.

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System Used: Windows

Assign a Unique Local IPv6 Address
Open Command Prompt as Administrator and run:

netsh interface ipv6 set address name="Wi-Fi" address=fd00:1:1::10

Configure IPv6 Address on Another LAN Device
Example address for the second device:

fd00:1:1::20

Test IPv6 Connectivity
From the first computer, run:

ping -6 fd00:1:1::20

The output should show:

Reply from fd00:1:1::20: time<1ms

Reply from fd00:1:1::20: time<1ms

Reply from fd00:1:1::20: time<1ms

Result: The unique local IPv6 address fd00:1:1::10 was configured successfully, and connectivity with the LAN device fd00:1:1::20 was verified using ping -6.
Let me know if you need help formatting these commands into a script or need the equivalent steps for a Linux or macOS system.

3. **Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.**
   Constraint: Document the steps and the observed address behavior in a text file.

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System Used: Windows

Enable Temporary IPv6 Addresses
Open Command Prompt as Administrator and run:

netsh interface ipv6 set privacy state=enabled

Check the Current IPv6 Address
Run:

Ipconfig

Example output:
Temporary IPv6 Address: 2001:db8:1::abcd

Disconnect and Reconnect the Network
Disconnect the Wi-Fi, reconnect it, and run:

Ipconfig

Check the Temporary IPv6 Address again.

Restart the Computer
After restarting, run:

Ipconfig

Check whether the temporary IPv6 address has changed.

Result: The temporary IPv6 address was configured and checked before and after network reconnection/reboot. The observed address can change because temporary IPv6 addresses are designed to provide privacy.

4. **Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.**

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```
restaurants = [
  {"name": "Pizza Hut", "ipv6": "2001:db8::1"},
  {"name": "Dominos", "ipv6": "2001:db8::2"},
  {"name": "Burger King", "ipv6": "2001:db8::3"},
  {"name": "McDonalds", "ipv6": "2001:db8::4"},
  {"name": "Subway", "ipv6": "2001:db8::5"}
]

search = input("Enter restaurant name: ").lower()

found = False

for restaurant in restaurants:
  if search in restaurant["name"].lower():
    print("Restaurant:", restaurant["name"])
    print("IPv6 Address:", restaurant["ipv6"])
    found = True

if not found:
  print("Restaurant not found.")
```

Example Output:

Enter restaurant name: pizza

Restaurant: Pizza Hut

IPv6 Address: 2001:db8::1

5. **Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.**

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IPv6 Subnetting Scenario and Configuration Steps

AI-Generated Scenario

An office network is assigned the documentation IPv6 prefix 2001:db8:abcd::/48. The network administrator divides it into /64 subnets for three departments:

1. Admin: 2001:db8:abcd:1::/64

2. Sales: 2001:db8:abcd:2::/64

3. IT Support: 2001:db8:abcd:3::/64

A sample host address for the Admin subnet is:

2001:db8:abcd:1::10/64

Note: 2001:db8::/32 is reserved for documentation and examples. These sample addresses are not intended for real internet connectivity.

Configuration Steps (Windows)

1. Open Command Prompt as Administrator.

2. Check the adapter name: netsh interface ipv6 show interfaces

3. Assign the sample IPv6 address to the primary adapter (replace "Wi-Fi" with the actual adapter name): netsh interface ipv6 add address interface="Wi-Fi" address=2001:db8:abcd:1::10

4. Verify the configuration:

ipconfig

or

ipconfig /all

5. Find the IPv6 Address under the chosen network adapter.

Example verification:

IPv6 Address: 2001:db8:abcd:1::10

Important:

This is a sample configuration for an assignment. The documentation prefix 2001:db8::/32 is not routable on the public internet. For a real network, use the IPv6 prefix assigned by your router, ISP, or network administrator. If the command fails or the address does not appear, confirm the adapter name and use an address/prefix appropriate for your network.